

researchers have found, and the person can be warned. Along the same lines, a security camera can feed a Bayesian learner that can then learn to identify suspicious persons at airports. For example, if someone gets out of a car and waits uncertainly for a long time before entering the terminal, it might mean a potential troublemaker.

Intel's robotics branch has an ongoing project that focuses on building Bayesian control systems that will help create more intelligent industrial robots. And robots are being taught, in Bayesian fashion, to navigate more effectively.

Bayesian inferences come in useful where individuals need to make strategic decisions of a certain kind repeatedly, for example, pricing policies, or military strategising. As the number of decisions made increases, Bayesian inferencing makes it easier and easier to make the next.

Bayesian is probably here to stay

Dr Marina Meila of the University of Washington says, "Bayesian inference is to statistics what the thermodynamical laws are to physics"—not fitting in nice-



<http://www.gametheory.net/Mike/applets/Bayes/Bayes.html>

A cool applet that demonstrates Bayes' Theorem. It enables you to answer the question, "How worried should I be if I tested positive for a particular disease?"

http://dmoz.org/Computers/Artificial_Intelligence/Belief_Networks/

A comprehensive collection of links to resources on Belief Nets, including BBNs.

<http://excalibur.brc.uconn.edu/~baynet/fieldedSystems.html>

This page lists some real-world systems that employ BBNs.

<http://yudkowsky.net/bayes/bayes.html>

An intuitive explanation of Bayesian reasoning, which provides a great non-technical understanding of how all things Bayesian work.

<http://crm114.sourceforge.net/>

All about the Controllable Regex Mutator, or CRM114.

ly with the rules, somewhat startling, but intuitive all the same.

It took centuries before Bayes' rule was conjoined with graphical methods to result in BBNs. And it's taken years for BBNs and Bayesian reasoning and learning to be taken as seriously as they are today. This is partly because probabilistic decision-making has been viewed as something dark and untrustworthy.

In the mid 90s, 'Fuzzy Logic' was much hyped; a phrase that was bandied about as the method of choice for almost any AI problem. Anyone talking loudly about fuzzy logic was looked up to and taken very seriously. Of course, fuzzy logic is employed today in several problems, but there was more hype about it than was warranted.

So are 'Bayesian Learning' and 'Bayesian methods' just buzzwords in today's AI and other scientific communities? Probably not. To quote Eric Horvitz again, "This is a foundation... it was overlooked for a while, but it is a foundation for reasoning." 

RAM MOHAN RAO

ram_mohan@thinkdigit.com

1/2 V AD